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Population physiologically based modeling of pirlimycin milk concentrations in dairy cows

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ABSTRACT

Predictions of drug residues in milk are critical in food protection and are a major consideration in the economics of treatment of mastitis in dairy cows. Nonlinear mixed-effects modeling (NLME) has been advocated as a suitable pharmaco-statistical method for the study of drug residues in milk. Recent developments in physiologically based pharmacokinetic (PBPK) modeling of intramammary drugs allow the combination of a mechanistic description of milk pharmacokinetics with NLME methods. The PBPK model was applied to NLME analysis of a data set consisting of milk drug concentrations from 78 healthy cows and 117 with clinical mastitis. Pirlimycin milk pharmacokinetics were adequately described by the model across the range of observed concentrations. Mastitis was characterized by increased variance in milk production volume. Udder residual volume was larger in cows with 1, or 2 or greater diseased mammary glands than in the healthy cows. Low-producing cows had a greater risk of prolonged milk residues. With the exclusion of the low-production cows, the model predicted that healthy cows required a milk discard time 12 h longer than that indicated by the label, and the diseased cows 36 h longer than indicated by the label. More pirlimycin was systemically absorbed in the gram-positive infected compared with the gram-negative infected or healthy cows, suggesting a greater risk of violative meat residues in gram-positive infected cows. Using NLME and PBPK models, we identified factors associated with changes in pirlimycin milk residues that may affect food safety. This model extends the verification of a simple physiologically based framework for the study of intramammary drugs.
Key words: pharmacokinetics, antibiotics, biostatistics, mastitis, mathematical modeling

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INTRODUCTION

Antimicrobial therapy is important for the management of bacterial mastitis in dairy cows but creates significant challenges in the control of drug residues in milk, which pose a hazard to food safety and production. The presence of antibiotic residues may cause direct effects on human consumers, but likely more important are interference with milk processing (Cogan et al., 2007) and contribution to antimicrobial resistance (Cerniglia et al., 2016; Van Vleck Pereira et al., 2016). Typical study designs for characterization of the persistence of drug residues in milk have used small cohorts of healthy cows and have defined milk discard times (**MDT**) based on extrapolation. These designs are unable to consider numerous factors contributing to variability in the pharmacokinetics of drugs in milk, such as disease state (Kissell et al., 2015) and production conditions (Whittem, 1999). Those factors may also contribute to the antibacterial effect, characterized by pharmacokinetic-pharmacodynamic (**PK-PD**) indices such as the time that drug concentrations exceed the pathogen MIC, which are minimally explored for intramammary drugs. Novel pharmaco-statistical approaches based on nonlinear mixed-effects models (**NLME**) have recently been applied to studies of meat residues, revealing food safety risks associated with disease (Wu et al., 2013a), route of administration (Leavens et al., 2014), and population variability (Wu et al., 2013b). These methods are also broadly applied for population inference for PK-PD parameters. However, the intramammary route differs greatly in pharmacokinetics compared with oral and parenteral routes (Whittem et al., 2012), which complicates the application of these methods. This suggests a role for suitable pharmacokinetic models that can accommodate the physiologic characteristics of the mammary gland.

In this study, a population pharmacokinetic NLME model of intramammary pirlimycin was developed using a physiologically based pharmacokinetic (**PBPK**) structural model based on those developed in our previous works (Whittem et al., 2012; Woodward and Whit-

tem, 2019b) and explored in others (Li et al., 2018). Data were drawn from 2 sources: a composite data set used for earlier development of a population model of pirlimycin residues (Whittem, 1999) and a field study of pirlimycin residues in cows with naturally occurring mastitis. A NLME was developed using the stochastic approximation expectation-maximization algorithm (**SAEM**; Lavielle and Mentré, 2007) as implemented in Monolix (Lixoft, France). This study focused on 2 primary objectives: first, to consider the effects of variability in field cases on the population distribution of MDT, including cows with mastitis, and second, to progress the PBPK model to include a drug with major known differences in both physicochemical characteristics and pharmacokinetics compared with those used previously.

METHODOLOGY

Sources of Data

Two primary data sources were used for this study. The complete data set included 195 cows, 1,190 milk volume data points, and 1,851 milk pirlimycin concentration data points, 600 of which were triplicate measures from 20 cows. Though the data were drawn from various studies, in all cases the concentration of pirlimycin in bucket milk (milk pooled from all mammary glands) was obtained using a microbiological assay, using *Micrococcus luteus* as the test organism (Friedlander and Moulin, 2004). The limit of quantification was 0.03 mg/L, lower than the U.S. Food and Drug Administration's tolerance of 0.4 mg/L (FDA, 2005) and the European Medicines Agency's maximum residue limit (**MRL**) of 0.1 mg/L (European Commission, 2010).

The first source (the regulatory data set) comprised composite data from various studies of pirlimycin pharmacokinetics conducted in the regulatory setting. Of the study cows, 78 were determined to be healthy at the time of study, whereas 26 had naturally occurring clinical mastitis. Briefly, this data set comprised the results of 5 studies conducted during product development that included a series of different dose regimens with intramammary administration, and a sixth field study with administration to cows with mastitis. Further details of the experiments that generated these data have been previously described in detail (Whittem, 1999). The second source (the Illinois data set) were from an experiment performed in 1999 at the research dairy of the University of Illinois, approved by the University's Institutional Animal Care and Use Committee. Ninety-one cows with clinical mastitis caused by gram-positive or gram-negative bacteria in one or more mammary glands were enrolled. The cows had visibly abnormal milk or a visibly inflamed gland(s). The affected gland(s) had a California Mastitis Test score of trace or greater, consistent with a high somatic cell count (Schalm and Noorlander, 1957). Foremilk was aseptically collected and cultured in duplicate on 5% blood agar and MacConkey medium. Sample collection, culturing, and pathogen identification were in accordance with procedures recommended by the National Mastitis Council (Hogan et al., 1999). After enrollment, 91 cows were treated with pirlimycin (Pirsue, Zoetis Inc., Parsippany-Troy Hills, NJ) by intramammary infusion. These cows were positive to the California Mastitis Test in one or more quarters. These cows received 50 to 200 mg of pirlimycin in each of 1 to 4 quarters on 1 to 4 dosing occasions, generally 24 h apart (i.e., dosing occasions followed by one milking without dosing). Each animal was studied on one occasion only. At each milking following drug administration, the bucket milk, combined from all mammary glands, was sampled for pirlimycin quantification, and the weight of milk produced was measured.

The total data for each animal included the volume of milk production at each milking, and the concentration of pirlimycin at each milking; covariates included the presence or absence of mastitis, and for cows with mastitis, the number of affected quarters, whether the infectious isolate was gram-positive or gram-negative. The characteristics of the complete data set are summarized in Table 1.

Due to the high systemic absorption of pirlimycin after intramammary dosing, it was anticipated that systemic pharmacokinetics of pirlimycin may contribute to the observed concentration-time relationship in milk. Data describing intravenous administration of radio-labeled pirlimycin were obtained (Friedlander and

Table 1. Summary of mastitis covariate characteristics for the pirlimycin-treated cows[1]

Characteristic	Source of data		
	Regulatory data set	Illinois data set	Total
Total number of cows	104	91	195
Mastitis			
Present	26	91	117
Absent	78	0	78
Gram-positive	10	54	64
Gram-negative	10	37	47
No gram-staining data	6	0	6
Number of diseased quarters			
0	77	0	77
1	27	85	112
2 or more	0	6	6

[1]Pooled data describing plasma concentrations are excluded.

Moulin, 2004). Data describing plasma concentrations of pirlimycin after intramammary administration were drawn from the same report (Friedlander and Moulin, 2004) comprising 16 data points. These are pooled data and the volumes of milk produced are not available; we assumed an arbitrary intermediate level of production (10 L/12 h).

Milk Production

The data set describes the individual-level production of milk at each time of sampling. This factor is an important contributor to the milk-level pharmacokinetics (Whittem, 1999). To explore this effect, a general linear mixed model was specified with milk volume as the response and random effects both for the individual and for the milking occasions within individuals. Various combinations of predictors were explored. Confidence intervals for the estimates, Akaike's information criterion (**AIC**), and the model residuals were examined to select the optimum model.

In the pharmacokinetic modeling, the volume of milk production from each milking was used directly in the model, instead of using a model-based prediction of milk volume as in earlier models (Whittem et al., 2012; Woodward and Whittem, 2019b). Where milk volume data points were missing from the data set, predictions for milk volume for these time points were generated, using conditional predictions from the general linear mixed model. It was assumed that rear mammary glands each contributed 29% of the total milk volume, and front mammary glands 21% (Ayadi et al., 2003).

Pharmacokinetic Model Specification

The pharmacokinetic model was adapted from a method developed for β-lactam antibiotics (Woodward and Whittem, 2019b), described in Figure 1. The final model comprises a systemic compartment that follows a standard 2-compartment model, having a central volume of distribution (V_c), drug elimination from the central compartment k_{40}, and exchange with a second compartment with rate constants k_{45} and k_{54}. A milk compartment is defined for each quarter. Drug exchange between the systemic compartment and milk is defined for each quarter, with plasma-milk elimination (k_{42}) of 5% of the systemic elimination rate, and milk-plasma elimination (k_{24}). Each quarter has a retention compartment with which drug is exchanged from milk, with rate constants k_{23} and k_{32}. The retention compartment represents a volume-independent mechanism for drug disposition during the intermilking interval, which is important for the total time course of drug concentration (Woodward and Whittem, 2019b). Doses are distributed throughout the quarters; where the data set included treatments of less than 4 quarters, treatments were assumed to have been administered to the rear quarters, as this detail was not available in the data set.

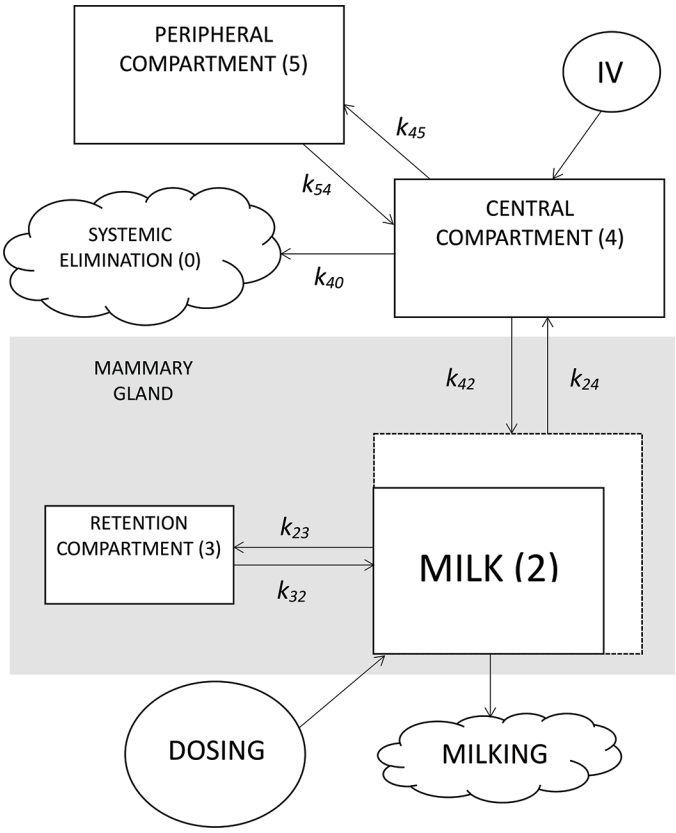

Figure 1. Schematic of the final structural model for pirlimycin. The gray region represents the mammary gland quarters, 4 of which are simultaneously modeled for each subject. Drug is administered by intramammary infusion (dosing) and removed at the time of milking, along with the current udder milk volume minus the residual volume. Within the quarter, drug is located either in milk (2), or in the retention compartment (3) from which it is not eliminated at the time of milking. Drug exchanges between the quarters and the central (systemic) compartment (4), and after systemic absorption follows a 2-compartment model (4 and 5). This model also describes intravenous pirlimycin administration (IV).

Independent estimates of the systemic pharmacokinetic parameters for pirlimycin were directly estimated from subjects treated with intravenous pirlimycin (Friedlander and Moulin, 2004). Tri-exponential models for these subjects were parameterized to the same ordinary differential equation (**ODE**) format as the physiologically based model; the mean estimates from these individuals were used as the fixed effects for k_{40}, k_{45}, and k_{54}, which were locked at these values in the subsequent models.

The same report (Friedlander and Moulin, 2004) describes the mean concentration of pirlimycin in plasma after intramammary administration of 200 mg

of pirlimycin per quarter at several time points within the intermilking interval. To incorporate these data, milk production from these cows was assumed to be the mean milk production from healthy cows in the overall data set, and the central compartment volume of distribution was estimated from the intravenously treated individuals. These data were incorporated into the final data set and contributed to the final mixed-effects model, with sampling from the central compartment and assuming the same error distribution as for the milk-concentration data. These data support the validity of the model of drug exchange between milk and plasma.

Implementation

As in the prior work (Woodward and Whittem, 2019b), the structural model was implemented in MATLAB. To improve the speed of computation, ODE for the intermilking interval were solved using the backward differentiation formulas in SUNDIALS (Hindmarsh et al., 2005).

The SAEM algorithm, as implemented in Monolix 4.3, was applied for maximum likelihood estimation of the model parameters. Custom code was written in MATLAB to define the structural model and to link to the Monolix software. In this study, 200 iterations of the Markov chain were initially used, and convergence of the parameter estimates was visually assessed for each analysis (Kuhn and Lavielle, 2005); where the convergence was not complete, further analysis was performed using up to 1,000 iterations. Single-chain analysis was used for this study.

For data below the analytical limit of quantification (**BLQ**; Woodward and Whittem, 2019a), the data points were defined as left-censored at the limit of quantification, 0.03 mg/L. All BLQ points were replaced in the data set with 0.03 mg/L and flagged as censored. The SAEM algorithm correctly uses the likelihood function for censored data (Samson et al., 2006). The total number of BLQ data points was 156 of 1,851 (8.4%). Considerations for analysis of BLQ data in the milk residue context have been recently reviewed, indicating that BLQ data of less than 10% of the data set have limited effect on parameter estimates where an NLME model is used (Henri et al., 2016).

Statistical Model Specification

This study used a conventional 3-level hierarchy of statistical effects, including fixed effects, random effects, and residual error (Sheiner et al., 1977). Random effects were at the level of interindividual variability. No interoccasional variability (Karlsson and Sheiner, 1993) components were included in this study. Parameters for each individual were therefore the sum of the fixed effects and one random effect.

A combination of forward and backward model building was used, with fixed and random effects, parameter covariance patterns, and covariates sequentially added or removed depending on their estimated importance in the model. Various combinations of normal and log-normal distributions for the parameters were evaluated. Constant, proportional, exponential, and combined (proportional + constant) error models were evaluated. These error models were iteratively tested in combination with various random effect models and covariates. The error model for the milk data was also used for the plasma data.

Model specifications were progressively adjusted to improve the model fit. The goodness-of-fit of candidate models was assessed using plots of the weighted residuals against observed concentration or sample time, identity plots of the observations and predictions, histograms of the parameter estimates and random effects, and evaluation of the relative standard error of the parameter estimates, eigenvalues, and residual error parameters. To distinguish candidate models, the −2 log-likelihood of the final parameter set was estimated using linearization, and the AIC was determined.

Internal Validation

The plots presented, assessment of the parameter estimates, and predicted characteristics of the population were generated using custom MATLAB functions.

Assessment of the quality of the parameter estimates of the final model was performed using the standardized visual predictive check; this method was selected rather than the visual predictive check, which was unsuitable due to the complex structure of the data set, with variations in dosing regimen and volume of milk produced, and presence of covariates (Wang and Zhang, 2012). Using estimated parameters for each individual, a Monte Carlo simulation was performed to generate a data series identical in structure to the original data set; 1,000 such simulated data points were generated for each observation, using the simulation function in Monolix. The predicted data points from the final model were then plotted against the time of observation by their percentile position in the simulated data. In this form, appropriate models have the predictions scattered randomly between 0 and 1 (Wang and Zhang, 2012).

Simulations

To examine the predictions of this model regarding MDT, a new data set of the same size as the pirlimycin data set was simulated, using the estimated individual parameters. Milk volume at each simulated milking was set to the observed median volume for the individual. The pharmacokinetic parameters were set to their individual estimates. The dose was set at the label dose of 50 mg twice at a 24-h interval, as indicated by the Food and Drug Administration-approved label for pirlimycin (FDA, 2007) and administered to all mammary glands. For each subject, the earliest milking predicted to have milk concentration below the MRL was identified (EMA, 2000). This individual-level time to MRL was used to assess and summarize the persistence of drug residues predicted by the final model.

RESULTS

Milk Production

The final general linear mixed model was defined as

$$\text{Volume} = \text{GPOS} + \text{GNEG} + (1|\text{ID}),$$

where GPOS and GNEG are categorical predictors for gram-positive or negative disease respectively, and 1|ID is the random effect for the individual level. The presence of gram-positive mastitis had a significant positive effect on milk production ($\beta = 2.82$, 95% CI = 1.05, 4.14), but gram-negative mastitis did not ($\beta = 0.387$, 95% CI = −1.06, 1.836) and was excluded from the final model. Neither gram-positive nor gram-negative mastitis had a significant effect on the interindividual variability (random-effect covariance), and both were excluded from the final model.

The presence of mastitis had an apparent effect on intraindividual variability; cows with mastitis were fitted more poorly by this model (Figure 2).

Model Development

Figure 1 describes the final structural model. This model has 2 milk drug eliminating processes: bulk elimination of the drug at milking and systemic drug absorption.

Systemic drug absorption was a necessary factor in this model. A series of attempts were made to estimate the systemic clearance, volume of distribution, and systemic udder clearance parameters, but incorporating the parameters determined after intravenous administration was more effective and facilitated further model development with greater algorithm stability. The absence of data describing plasma concentrations for the intramammary treated cows prevented meaningful estimation of interindividual variability of these parameters, so their true variance contributes to the residual error in this model.

A proportional error model was selected for the drug concentration variable. Constant error models showed a poor fit compared with the proportional model, and after testing of a combined proportional-constant error models the constant parameter was too small to be of importance and was discarded. The residual error parameter (b = 0.302) was estimated precisely (relative SE 2%).

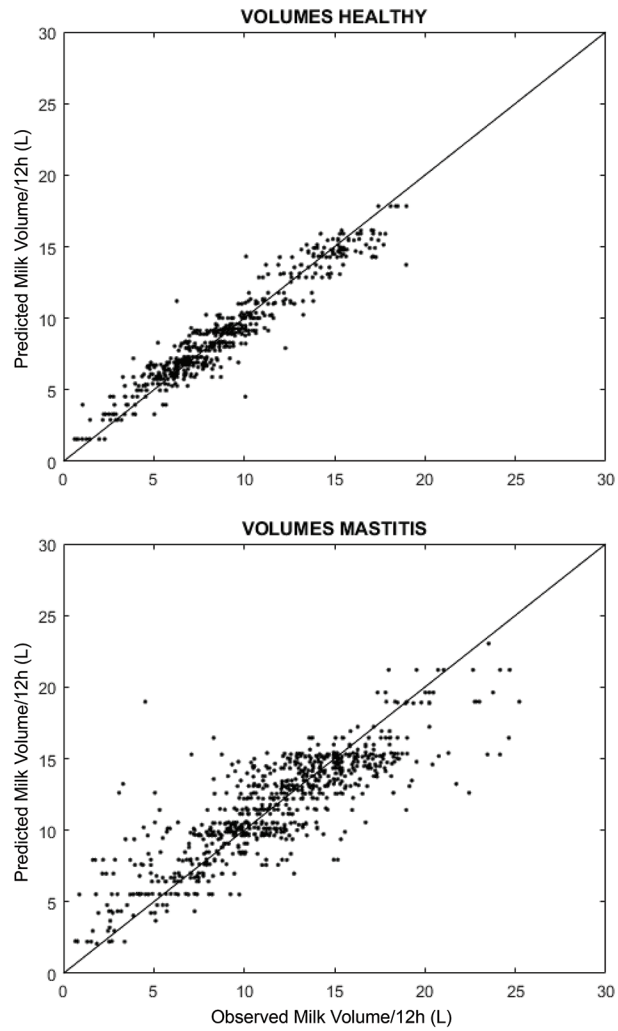

Figure 2. Identity plots for the observed (x-axis) and predicted (y-axis) volumes of milk production per 12 h, for healthy cows in the upper panel and cows with mastitis in the lower panel, where the solid line is the line of identity (y = x).

Table 2. The estimated pharmacokinetic parameters, divided based on the categorical covariates[1]

Parameter	Covariate	Fixed effect	*P*-value	Random effect
R	Diseased quarters = 0	0.993 (11%)	*	0.584 (15%)
	Diseased quarters = 1	1.99 (12%)	$7.8*10^{-8}$	0.825 (13%)
		β = 0.838 (19%)		
	Diseased quarters = 2	4.73 (38%)	$6.4*10^{-5}$	
		β = 1.58 (25%)		
k_{24}	Gram-positive = 0	0.0568 (5%)	*	0.437 (12%)
	Gram-positive = 1	0.0721 (8%)	0.0095	0.573 (9%)
		β = 0.237 (39%)		
k_{23}		0.00418 (10%)		0.725 (12%)
k_{32}	Mastitis = 0	0.107 (11%)	*	0.663 (10%)
	Mastitis = 1	0.0553 (11%)	$3.4*10^{-8}$	
		β = −0.699 (21%)		

[1]All parameters are log-normally distributed, where fixed effect is the distribution median, and random effect is the SD. Bracketed values are the relative SE of the estimate (percentage). β indicates the effect size of the categorical covariates.

[2]*P*-values are from the Wald test where the test hypothesis is β = 0. *indicates the base effect.

Parameter Estimates

Fixed effects for the residual volume (0.993 L) and the systemic absorption rate (0.0556 h^{-1}) were precisely estimated (RSE 11 and 5%, respectively). Fixed effects of the parameters associated with the retention compartment, the milk to retention compartment rate constant k_{23} (0.0042 h^{-1}) and retention compartment to milk rate constant k_{32} (0.107 h^{-1}) were also well estimated (RSE 10 and 11%, respectively).

The presence of disease had significant effects on the model parameters. The rate of transfer from the retention compartment to milk was ($P < 10^{-7}$) smaller (β = −0.699) for the mastitis cohort than for the healthy cohort. Evaluation of Gram staining as a covariate on this parameter did not reveal a clear effect. Gram-positive disease weakly increased ($P = 0.0095$) the systemic absorption rate (β = 0.227).

For the residual volume parameter R, a significant effect was associated with disease. Animals with one diseased mammary gland had greater ($P < 10^{-7}$) residual volume (β = 0.838) than healthy cows. Those with 2 or more diseased quarters also had greater ($P < 10^{-4}$) residual volume than the healthy cows (β = 1.58). Effects for animals with 2 or 3 diseased glands were pooled, as the number of subjects with 2 or more diseased glands (n = 6) was too small to obtain a stable estimate. For the animals with 2 or more diseased glands the RSE of the covariate effect was larger (RSE = 39%), the weakest parameter estimate in the final model.

Alveolar transfer parameter, representing the proportion of drug formulation delivered to the alveolar compartment, was removed from the final model as estimates of the parameter were high (fixed effect approximately 1) and minimally variable, indicating extensive and rapid formulation distribution. This parameter was locked at 1 for all individuals, with no apparent effect on the final model's goodness-of-fit.

Random effects were precisely estimated with RSE < 20% (Table 2). The presence of mastitis was tested as a covariate on the random effect covariance; the diseased cohort had higher variability for R and k_{24} (Table 2).

Internal Validation

The goodness-of-fit indicators demonstrated that the final model was a suitable description of the data. The fit of the fixed effects model only, although substantially poorer than the model with the random effects, showed no evidence of bias. Estimation of the random effects substantially improved the quality of the predictions, though there was evidence of weak bias toward overprediction (Figure 3). The overpredictions were associated with a small number of data points, some of which were severely overpredicted. The standardized visual predictive check (Supplemental Figure S1, https://doi.org/10.3168/jds.2020-18760) also indicated a trend toward overprediction, with bias evident in the position of the median line.

Assessment of the individual-level concentration-time predictions demonstrated that severe errors, including some concentrations overpredicted by 5 times or more (Supplemental Figure S2, https://doi.org/10.3168/jds.2020-18760), were predominantly associated with sample time points after dosing, occurring in 12 subjects, form both the healthy and diseased cohorts. Compared with data from other subjects receiving similar dose regimens, the measured concentrations were about an order of magnitude lower. No values of the parameters achieved a closer fit to these data points. Review of the individual-level independent variables and covariates revealed no explanatory factors for these errors.

There were some apparent correlations between the random effects. Residual volume was positively correlated with mean volume of production in the healthy cows (R = 0.284, $P = 0.0122$) but not mastitis cows (R = 0.0421, $P = 0.651$). There were substantial correlations between the retention compartment parameters (k_{23} and k_{32}). The systemic absorption rate (k_{24}) had no apparent correlations.

Simulations

The simulated data set was generated to evaluate milk residues at the individual level from the observed cows, based on the label dose regimen, correcting for diversity in the dose regimen applied to many of the treated cows. For the 12 cows with at least 1 observed data point overpredicted by a factor of 5 or more in the model (see the Internal Validation section), simulation results are presented first with those cows included, and second with these cows removed from the data set. The individual-level number of milkings to reach MRL are shown in Supplemental Figure S3 (https://doi.org/10.3168/jds.2020-18760; excluding outliers).

For the healthy cohort, 14 of 78 (17.9%) individuals had predicted MDT that exceeded the labeled milk discard time of 36 h, with 9 of those cows having milk production exceeding 5 L/12 h (Supplemental Figure S3). Two of these violative healthy cows had least one poorly fitted, overpredicted data point, including one individual with prolonged MDT and greater than 5 L/12 h milk production; these individuals were then excluded from analysis (Supplemental Figure S3).

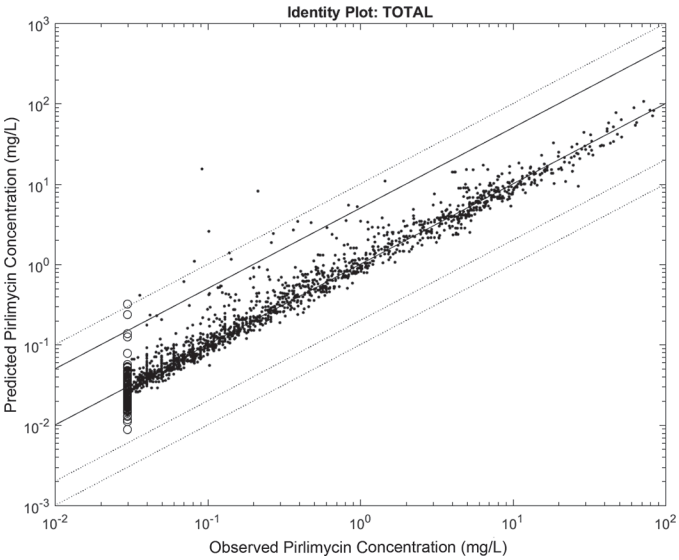

Figure 3. Identity plot for all cows, where the central solid line is the line of identity (y = x). Open circles are data points below the analytical limit of quantification (0.03 mg/L). Outer lines represent error of 5× and 10×, respectively, above and below the line of identity.

The estimated MDT from the model predictions for the healthy subjects was therefore sensitive both to the selection of the acceptable minimum production volume, and the criterion for the removal of poorly fitted subjects. After exclusion of these individuals, the healthy cohort failed to conform to the labeled 36-h MDT, because 8 of 78 cows (10.3%) were first below the MRL at milking 4 after the last dose. Longer residue persistence was associated with lower-producing cows, with the slowest eliminator requiring 6 milkings to reach the MRL. Therefore, excluding 3 low producers, the MDT indicated for this cohort was 4 milkings, 1 milking longer than that indicated by the label.

In the gram-positive disease cohort, 3 cows of 64 (4.7%) were identified that had residues exceeding the MRL at the labeled milk discard time. These subjects were not low-producing and were not excluded as containing outlier data points. The MDT of 6 milkings was indicated for the gram-negative cohort.

For the gram-negative disease cohort, 13 individuals were identified with residues persisting after the labeled MRL; 7 of these individuals had production lower than 5 L/12 h. Of these 13 cows, 6 had milk production exceeding 5 L/12 h. None of these individuals had any outlier data points. The calculated MDT for the gram-negative infection subjects was therefore sensitive to the exclusion of low-producing cows and to outlying individuals. The MDT of 6 milkings was also indicated to ensure residue control for these cows.

Overall, the number of diseased quarters was minimally associated with predicted individual-level MDT (Supplemental Figure S3). Though an increased residual volume was associated with the number of affected quarters (Table 2), contributing to greater drug retention, this did not result in a marked change in the observed outcome for this cohort.

DISCUSSION

This study combined a novel PBPK model of the mammary gland with an advanced NLME approach for the analysis of a large data set. The study is comparable in size, based on subject and sample numbers, to the largest NLME in veterinary applications, including an investigation of amoxicillin breakpoints in pigs (Rey et al., 2014) and mavacoxib dose regimens in dogs (Cox et al., 2011). In the study of drug residues, large sample populations that include outlying individuals are essential to confidently estimate the underlying population distributions from which withholding times can be reli-

ably determined (Concordet and Toutain, 1997). The outcome of this study is therefore of critical interest both as a description of population-level variability in milk residue elimination for pirlimycin, and as an exploration of the various assumptions underlying methods supported by regulatory guidelines, which are based on extrapolations from small numbers of subjects. This method theoretically improves the external validity of model-based findings compared with previous NLME studies of milk residues (Whittem, 1999).

Quality of the final model predictions, illustrated for all cows in Figure 2, demonstrates that the model was an adequate description of the pharmacokinetics of pirlimycin. After intramammary administration, a large proportion of pirlimycin mass is systemically absorbed, up to 50% (Friedlander and Moulin, 2004), in contrast to the β-lactam drugs used in earlier model building (Ziv and Sulman, 1975). Further, approximately 5% is excreted in milk after intravenous administration (Friedlander and Moulin, 2004). Incorporating these observations as parameters in the model allowed these important processes to be included, as they could not be determined from the milk concentration data alone. The structural model developed using data from β-lactam drugs was suitable in this study for pirlimycin; this demonstrates the suitability of the proposed structural model as a generalized description of intramammary pharmacokinetics, including in cows with mastitis, and also its utility in incorporating relevant prior information.

Due to the apparent importance of intermilking variability in milk production, especially in the mastitis cows (Figure 2) and the constant 12-h milking frequency, milk volumes were drawn directly from the data rather than predicted throughout the intermilking interval. This simplified the structural model compared with the β-lactam model (Woodward and Whittem, 2019b). As all samples from the data set were taken at 12-h intervals, the data were insufficient to identify the variety of milk volume-associated parameters that were previously described, as the nonlinear function for milk volume cannot be determined with only one time point. A possible alternate approach was to estimate interoccasion variability with the milking defined as the occasion variable, but this greatly increased the complexity and was considered infeasible. In the absence of the cistern compartment, assuming homogeneity of drug throughout the milk within each mammary gland, the model is still physiologically reasonable, but prediction of pirlimycin residues outside of 12-h milking intervals would require additional data to predict reasonable milk volumes at these times, and to assess the additional effect of cistern volume (Woodward and Whittem, 2019b). Early investigation in the present study demonstrated that the alveolar transfer effect for pirlimycin was close to 1, indicating relatively complete formulation distribution. The absence of this process would be expected to cause underestimation in the initial concentrations after milking, but this was not noted (Figure 3).

The final model was a close description of the data across the entire concentration range of the observations. However, for several individuals the model significantly overestimated some milk drug concentrations (Figure 3). These occurred in both healthy and mastitis cohorts. None of the tested error models or covariates improved goodness-of-fit for these data points. As many of these over-estimates were associated with samples immediately following dosing, one explanation is that they may be associated with operator error in drug administration. An alternate explanation, that the quality of the administered drug preparation may be involved, is unlikely as a registered drug product from a reputable company was used. The similar time and direction of these errors suggests that a common unknown mechanism is responsible. In this case of significant overestimation, severe violation of the assumption of homogeneity of drug throughout the compartment, so that the sample was drawn from a low-concentration region, is an appealing explanation. Further inference about the nature of this process is not feasible with currently available data. If these observations are associated with genuinely low milk pirlimycin concentrations, those individuals would have a lower time that drug concentrations exceed the pathogen MIC and represent a cohort more likely to experience treatment failure (McKellar et al., 2004).

Systemic pharmacokinetics of pirlimycin after intravenous administration are characterized by slow elimination, substantial compartmental behavior, and a small amount of transfer to milk (Friedlander and Moulin, 2004). The available plasma pirlimycin data demonstrated that first-order transfer of pirlimycin from milk was a reasonable approximation, though this assessment was impaired by the unavailability of raw data for these subjects. These effects contributed to predicted milk concentrations of pirlimycin but did not adequately describe observed drug retention in the terminal phase.

An apparent increased level of production associated with gram-positive infections is not consistent with previous reports on the effects of mastitis on production level (Hertl et al., 2014); however, as the data only indicate the level of production during the treatment period, it is plausible that the apparent positive effect on production associated with gram-positive disease indicated predisposition of higher-production cows to gram-positive infections, rather than an effect of dis-

ease. As the cows were drawn from several herds and multiple breeds are represented, the predicted effects of disease in this model are probably confounded. However, this model of milk volume is of descriptive importance only, as direct incorporation of the milk volume data as a predictor variable for the pharmacokinetic model prevented this uncertainty from influencing the milk-concentration predictions. Both gram-positive and gram-negative disease were associated with increased within-individual variability in milk production, which is of substantial importance given the influence of milk volume on drug concentrations, particularly soon after dosing. This effect will contribute greater uncertainty of predictions for diseased cows compared with healthy cows. As current methods define MDT for outlier individuals based on predicted milk drug concentration, the elevated within-individual variability of diseased cows might reduce the external validity of residue studies conducted in healthy cows, especially when low volumes of milk are produced. The current data set does not describe the time course of disease; as some diseased cows may have recovered during the study and their within-individual variability returned to normal, the true effect of disease on within-individual variability may be underestimated. A systematic understanding of production and variability in treated populations could be an important contributor to accurate population predictions.

The importance of the retention compartment indicates that more pirlimycin is retained in the mammary gland than is predicted by the combination of udder residual volume and redistribution of pirlimycin from plasma. In the absence of this compartment (i.e., $k_{23} = 0$), there was increased variability around the model predictions, greater residual error, and the residual volume parameter was larger. Unlike the residual volume, or exchange of drug with the systemic compartment, the existence of this effect is inferred from the data, rather than from knowledge of the physiology, and the real physical process it represents is unknown. Binding of drug to udder tissue occurs to a variable extent between drugs (Ziv et al., 1976). Other possible sites for drug binding include protein and lipid components of milk, which are heterogeneously distributed throughout the gland (Ayadi et al., 2004) and also differ between healthy and diseased mammary glands (Bruckmaier et al., 2004). In this case, where the drug formulation is an aqueous gel, there is limited possible ongoing interaction between drug in milk and remaining formulation components. For lipid-based drug formulations, ongoing drug formulation interactions might contribute to this process, and this may include delayed-release characteristics of these formulations. A priority for further mechanistic studies of the pharmacokinetics of the mammary gland would be to identify and measure physical processes associated with drug retention.

The retention compartment is proposed as the major drug-specific effect contributing to the observed concentration-time relationship. In the absence of the retention compartment, the amount of drug retained at each milking is concentration-proportionate, and the duration of drug persistence would be dependent only on cow and production factors and the dose regimen. However, drugs differ greatly in their elimination rate to an extent that cannot be accounted for by these factors alone; for example, despite a relatively fast plasma elimination half-life after intravenous administration, tilmicosin is retained in the mammary gland for extended periods after parenteral (Ziv et al., 1995) or intramammary (Smith et al., 2009) administration. These factors demonstrate the utility of this modeling approach in allowing the disentanglement of cow, drug, and management factors. An experiment in which multiple drugs are simultaneously administered to the same quarter, and model parameters estimated from all at the same time, would be a viable approach to directly identify drug factors.

The presence of gram-positive disease resulted in an increased rate of systemic drug absorption (k_{23}) from the mammary gland (Table 2). Though the effect size was small ($\beta = 0.237$), it was clearly evident ($P = 0.0095$) and its estimate was precise (RSE 39%). No similar effect was evident for cows with non-gram-negative infections. For the other parameters, the Gram-staining effect was not significant in the final model and the presence of disease was used as the covariate instead. Altered rate of drug absorption from the gland suggests disruption of the blood–milk barrier, as absorption for pirlimycin likely occurs by simple diffusion (Ziv and Sulman, 1975). Lipopolysaccharide results in direct effects on the integrity of blood–milk barrier tight junctions by disruption of claudins (Kobayashi et al., 2013), but would be expected to cause at least the same effect in gram-negative infections. Further effects on vascular endothelial function are associated with a variety of inflammatory mediators which contribute to formation of leukotrienes and prostaglandins in the gland (Piotrowska-Tomala et al., 2015). In this model, effects of disease on the plasma-milk clearance rate were not estimated, as plasma data were not available. It is therefore unclear if the proposed disruption of the blood–milk barrier results in increased drug diffusion in both directions. Increased plasma exposure could also result in elevated meat withholding time for pirlimycin in cows with mastitis; its meat withholding time is relatively long (21 d). This is particularly important for the gram-positive disease cohort, as pirlimycin is only effective for gram-positive infections (Thornsberry et

al., 1993); treatment of gram-negative infections in this study occurred before culture results were available.

The number of mammary glands, modeled as a categorical covariate, influenced the udder residual volume (Table 2), and this covariate was supported by the AIC. The data set contained low diversity for this variable, as all but 6 of the cows had either zero or one affected glands, resulting in limited power. However, the data indicated that those individuals with 2 or more diseased glands had elevated residual volume compared with those with one diseased gland, which had greater residual volume than the healthy cows (Table 2). This indicates greater drug retention at the time of milking, with an increasing number of affected glands. This ordering of the covariate estimates suggests proportionately greater retention with increasing numbers of affected quarters; however, the effect of 2 or greater diseased quarters was associated with weaker confidence (RSE 39%) limited by the small number of cows.

Effects of disease on residual volume may indicate a truly decreased efficiency of milking, with the physical volume of milk remaining in the gland being larger in diseased than healthy glands. This effect is plausible if milk flow throughout the gland is impeded due to accumulation of inflammatory material or changes in milk composition (Bruickmaier et al., 2004), or if pain or discomfort associated with mastitis causes difficulty in machine milking (Sepúlveda-Varas et al., 2016). Though the somatic cell count and the peak milk flow and over-milking duration are associated, the direction of causality is not clear (Tančin et al., 2007), and no studies are identified that have specifically measured udder residual volumes in cows with mastitis compared with healthy controls. In this case, the observed effects of disease on residual volume would be expected to be drug-independent, and contribute to greater retention of any drug at milking. However, changes in the composition of residual milk in disease (Bruickmaier et al., 2004) may result in changes in drug binding characteristics in the residual milk, causing an apparent increase in volume without changes in real physical volume. Evaluation of simultaneous pharmacokinetics of multiple intramammary drugs in diseased and healthy cows would be an effective method to identify this mechanism, particularly if combined with direct measurement of the residual volume and its drug concentration using supra-physiologic oxytocin (Thompson et al., 1973).

Effects of disease on the pharmacokinetic parameters were detected, indicating that a healthy experimental cohort is likely to be a poor model of the pharmacokinetics of pirlimycin in diseased cows in the field. However, the overall effect on the time course of milk drug concentrations was relatively small. This study therefore did not support that the diseased cows were of elevated residue risk at the label MDT than the healthy cows. However, the variability in pharmacokinetics in the treated population indicates clearly that small experimental studies are likely to result in a poor estimate of the range of drug concentrations at the individual animal level. Though this study considered only individual-level assessments of the times to reach MRL, population-level estimation of MDT was beyond the scope of this study and is considered in a subsequent analysis of a subset of the Illinois data set (in preparation).

Consideration of the time above MIC, the efficacy-linked index for the lincosamides (McKellar et al., 2004) was not directly possible from this model as the milk volume at times during the intermilking interval was not estimated; however, as the volume increases during the intermilking interval, resulting in progressive dilution (Whittem et al., 2012) the predicted concentrations at milking are potentially a useful indicator. Further assessment of the pharmacokinetics-pharmacodynamics of pirlimycin in the mammary gland would require data from additional milking times to parameterize the full milk volume model.

This model demonstrates that for the studied population, individual cows with clinical mastitis required an elevated MDT (48–72 h) to achieve the standard residue target (0.4 mg/L), whether infection was gram-positive or gram-negative. Several heathy cows also failed to conform to the labeled MDT (36 h), and these predictions are sensitive to the exclusion of low-producing cows, which had greatly elevated risk of violative milk concentrations at the individual level across all disease classes. An increased risk of violative milk concentrations at the individual level was therefore associated with the variability in the treated population, and cows producing low volumes of milk. Regulatory agencies should consider these factors in their approach to milk residues management.

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may be provided to readers upon written request to A. Woodward.

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